

Ethan Ernesto Lopez

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TECHNICAL SKILLS

Statistical Tools & Languages – SQL | Python | R | C++ | Java | Git | GitHub | Docker | Linux | Unix | Excel | Matplotlib | Dyplr
Libraries & Platforms – Pandas | NumPy | Scikit-Learn | Jupyter Notebook | Google Colab | Alteryx Cloud | Tableau | Docker
Data – Hypothesis Testing | EDA | Regression | Clustering | Machine Learning | Social Network Analysis | Classification Models

PROFESSIONAL EXPERIENCE

Research Assistant – Social Network Analysis (MR-QAP) | *Chapman University* March 2026 – Present

- Evaluate collaboration network models for 240+ first-year STEM students in 11 classrooms via Python and MRQAP.
- Enhance datasets in cleaning and transformation methods, improving analytical precision and validation by 20%.
- Apply 4+ statistical methods including MR-QAP, ANOVA, and density to assess team stability and quality interactions.
- Develop more than 7 network visualizations and graphs explaining 64.4% variation in team formation outcomes.

Volunteer -> Senior Leadership Instructor | *UTO Taekwondo – Cypress, CA* March 2018 – Present

- Began as a volunteer and transitioned into a paid role, providing office support and handling customer inquiries.
- Supervise 15+ youth per shift in self-defense, manage entry-level team members, and oversee studio operations.

Academic Tutor (Student Tutor) | *Chapman University* September 2024 – May 2026

- Tutored 5+ students quarterly for 10 STEM and business topics in statistics, data analysis, and quantitative modeling.
- Expanded on techniques in regression, hypothesis testing, and data interpretation to improve problem-solving skills.
- Utilized over 13 statistical procedures in Python (NumPy, Pandas), R, and Excel, analyzing student feedback metrics.

Data Entry/Admin Assistant – Office of Accreditation & Assessment | *Chapman University* June 2025 – August 2025

- Maintained accuracy in graduate databases with 50+ students, validating survey metrics in accreditation assessments.
- Conducted research and enrollment trend analysis on more than 70 alumni to support counseling program outcomes.
- Leveraged visualizations using Excel and Qualtrics to improve stakeholder understanding of internships by 25%.

Teaching Assistant – Summer Engineering Academy (Arduino Cars) | *Chapman University* June 2025 – July 2025

- Provided technical assistance to 40+ students in Arduino programming, circuit design, and robot hardware assembly.
- Guided workshops in autonomous robot vehicle debugging methods to ensure 80% deployment success per section.
- Managed more than 20 materials and maintained documentation using Office tools to support program operations.

RELEVANT PROJECTS & EXPERIENCE

Spotify Tune Success Analytics Dashboard – *Business Intelligence & Data Visualization* April 2026 – May 2026

- Conducted analysis on over 130,000 Spotify songs with 17 audio metrics to identify key factors in streaming success.
- Built 10+ interactive Tableau visuals, enabling users to gain insights for 5+ correlations in audio song performances.
- Applied methods in hypothesis tests to reveal over 6 significant song popularity variables (energy, danceability, etc.).

Predicting Physical Activity From Wearable Health Metrics – *Machine Learning* April 2026 – May 2026

- Performed data analysis on over 6,000 health observations, observing physiological predictors for physical activities.
- Built machine learning workflows for data preprocessing, tuning 6+ hyperparameters for classification algorithms.
- Achieved an ROC-AUC of 0.92 with 76% model accuracy, feature importance analysis identifying 7 key classifiers.

Large-Scale Customer Segmentation Analysis – *Big Business Predictive Modeling* November 2025 – December 2025

- Collaborated with 2 others to integrate Jupyter/Colab machine learning with over 9 million online booking records.
- Implemented feature engineering and data transformation to apply clustering analysis for 6 customer segmentations.
- Trained 5+ regression and classification models, designing Matplotlib visuals to present to a non-technical audience.

EDUCATION

Chapman University, Orange, CA

Bachelor of Science in Data Science, Summa Cum Laude
Minor in Business Administration

May 2026
Program Honors Graduate
3.92 GPA

ACADEMIC AWARDS AND LEADERSHIP

Fowler School of Engineering Outstanding Senior Award – Data Science

2026

Outstanding Tutor Certificate of Appreciation

2026

Presidential Scholarship Recipient

2022–2026

Provost's List

2022–2025